

Vector Calculus

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1 Euclidean space

1.1 Basic Concepts

We define the **Euclidean space** $\mathbb{R}^n$, $n \geq 1$, as the set of all elements in $\mathbb{R}$:

$$x = (x_1, x_2, \dots, x_n) \quad \text{where } x_i \in \mathbb{R} \text{ for } i = 1, 2, \dots, n.$$

The **Standard basis** of $\mathbb{R}^n$ is given by the vectors:

$$e_1 = (1, 0, \dots, 0), \quad e_2 = (0, 1, \dots, 0), \quad \dots, \quad e_n = (0, 0, \dots, 1).$$

Then, any vector $x \in \mathbb{R}^n$ can be expressed as a linear combination of the standard basis vectors:

$$x = x_1 e_1 + x_2 e_2 + \dots + x_n e_n.$$

1.2 Properties of Euclidean Space

The Euclidean space $\mathbb{R}^n$ has the following properties:

- **Addition:** For any $x, y \in \mathbb{R}^n$, the sum $x + y$ is defined component-wise:

$$x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n).$$

- **Scalar Multiplication:** For any scalar $\alpha \in \mathbb{R}$ and vector $x \in \mathbb{R}^n$, the product αx is defined as:

$$\alpha x = (\alpha x_1, \alpha x_2, \dots, \alpha x_n).$$

- **Scalar Distributivity:** For any scalars $\alpha, \beta \in \mathbb{R}$ and vector $x \in \mathbb{R}^n$:

$$(\alpha + \beta)x = \alpha x + \beta x.$$

- **Vector Distributivity:** For any scalar $\alpha \in \mathbb{R}$ and vectors $x, y \in \mathbb{R}^n$:

$$\alpha(x + y) = \alpha x + \alpha y.$$

- **Scalar Commutativity:** For any scalars $\alpha, \beta \in \mathbb{R}$ and vector $x \in \mathbb{R}^n$:

$$\alpha(\beta x) = (\alpha\beta)x.$$

- **Multiplicative Identity:** For any vector $x \in \mathbb{R}^n$:

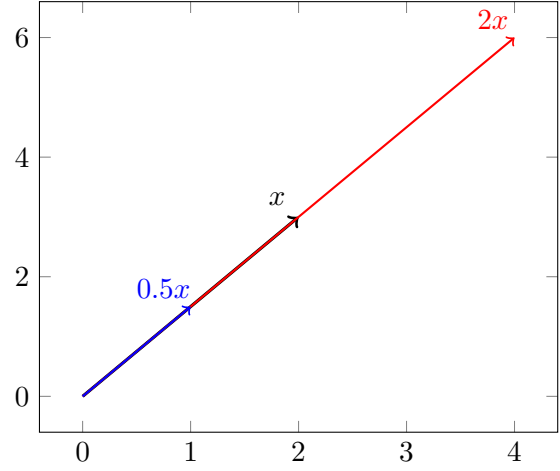
$$1 \cdot x = x.$$

- **Multiplicative Zero:** For any vector $x \in \mathbb{R}^n$:

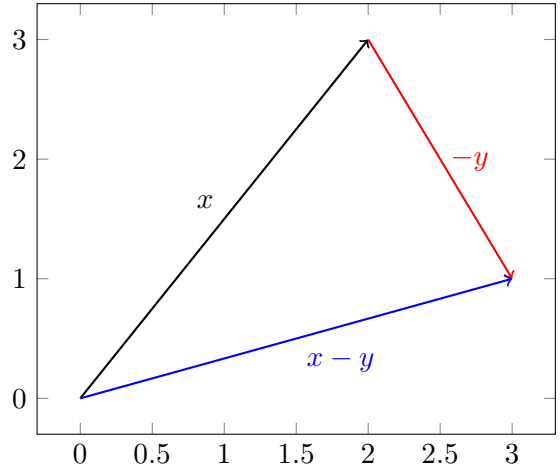
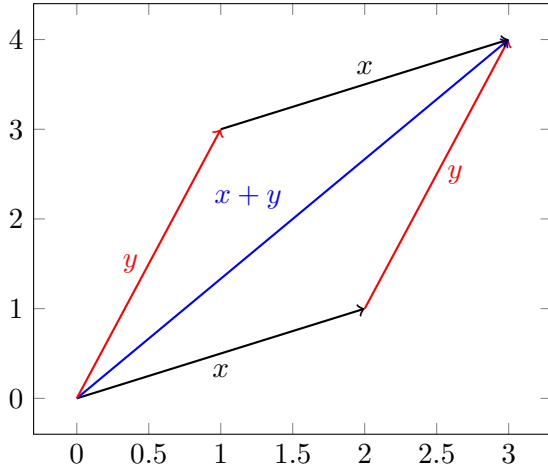
$$0 \cdot x = 0.$$

1.3 Geometric Interpretation

The Euclidean space $\mathbb{R}^n$ can be visualized as an n -dimensional space where each point corresponds to a vector. Given a vector $x = (x_1, x_2, \dots, x_n)$, and a scalar $\alpha \in \mathbb{R}$, the vector αx represents a point that is scaled by the factor α in the direction of x :



Given two vectors $x, y \in \mathbb{R}^n$, their sum $x + y$ represents the vector obtained by placing the tail of y at the head of x . Similarly, the difference $x - y$ can be visualized by reversing the direction of y and adding it to x .



1.4 Norm and Distance

The Euclidean space is a **normed vector space** with the norm defined as:

$$\begin{aligned} \|\cdot\| : \mathbb{R}^n &\rightarrow \mathbb{R} \\ x &\mapsto \|x\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}. \end{aligned}$$

The norm satisfies the following properties for all $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$:

- **Non-negativity:** $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = 0$.
- **Scalar Multiplication:** $\|\alpha x\| = |\alpha| \|x\|$.
- **Triangle Inequality:** $\|x + y\| \leq \|x\| + \|y\|$.

Remark

Associated with the norm is the **distance function** $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined by:

$$d(x, y) = \|x - y\|.$$

The distance function satisfies the following properties for all $x, y, z \in \mathbb{R}^n$:

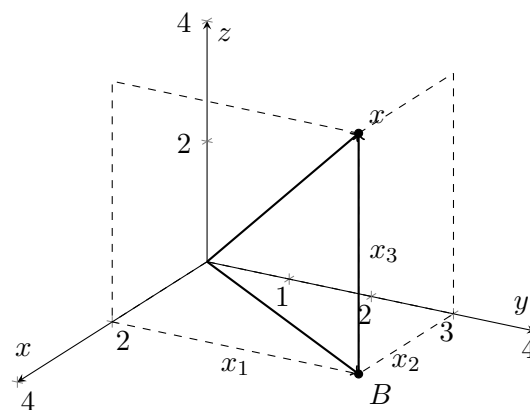
- **Non-negativity:** $d(x, y) \geq 0$ and $d(x, y) = 0$ if and only if $x = y$.
- **Symmetry:** $d(x, y) = d(y, x)$.
- **Triangle Inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

1.4.1 Norms in $\mathbb{R}^3$

In $\mathbb{R}^3$, the norm of a vector $x = (x_1, x_2, x_3)$ is given by:

$$\|x\| = d(O, x) = \sqrt{x_1^2 + x_2^2 + x_3^2},$$

where O is the origin and B is the projection of x onto the xy -plane:



1.5 Inner or Scalar Product

Let $x, y \in \mathbb{R}^n$. The **inner product** (or **scalar product**) of x and y is defined as:

$$x \cdot y = \langle x, y \rangle = x_1 y_1 + x_2 y_2 + \dots + x_n y_n.$$

The inner product satisfies the following properties for all $x, y, z \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$:

- **Positive Definite:** $x \cdot x \geq 0$ and $x \cdot x = 0$ if and only if $x = 0$.
- **Symmetry:** $x \cdot y = y \cdot x$.
- **Bilinear Form:** $\langle \lambda x + \mu y, z \rangle = \lambda \langle x, z \rangle + \mu \langle y, z \rangle$ for all $\lambda, \mu \in \mathbb{R}$.

1.6 Cauchy-Schwarz Inequality

For any vectors $x, y \in \mathbb{R}^n$, the Cauchy-Schwarz inequality states that:

$$|x \cdot y| \leq \|x\| \|y\|.$$

Equality holds if and only if x and y are linearly dependent.

Proof. If $y = \lambda x$ for some $\lambda \in \mathbb{R}$, then:

$$|x \cdot y| = |x \cdot (\lambda x)| = |\lambda| \|x\|^2 = \|x\| \|\lambda x\|.$$

If x and y are linearly independent, consider the vector $z = \lambda x + y$ for some $\lambda \in \mathbb{R}$. Then:

$$0 \leq \langle z, z \rangle = \|z\|^2 = \langle \lambda x + y, \lambda x + y \rangle.$$

Expanding this, we have:

$$\|z\|^2 = \lambda^2 \|x\|^2 + 2\lambda \langle x, y \rangle + \|y\|^2,$$

which is a quadratic function in λ . For $f(\lambda) \geq 0$ for all λ , the discriminant must be non-positive:

$$(2 \langle x, y \rangle)^2 - 4 \|x\|^2 \|y\|^2 \leq 0.$$

Simplifying, we get:

$$4 \langle x, y \rangle^2 \leq 4 \|x\|^2 \|y\|^2,$$

which leads to:

$$|x \cdot y| \leq \|x\| \|y\|.$$

□

1.7 Theorem: Inner Product and Norm

For any vectors $x, y \in \mathbb{R}^n$, the following relation holds:

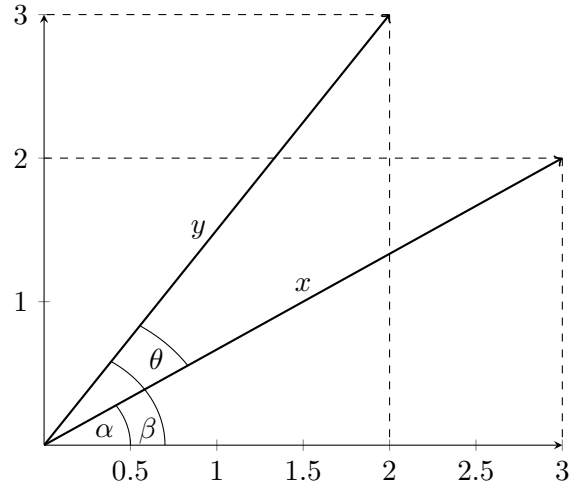
$$\langle x, y \rangle = \|x\| \|y\| \cos \theta,$$

where θ is the angle between the vectors x and y .

Proof. Consider the vectors x and y in $\mathbb{R}^2$. Let the angle between x and the positive x -axis be α , and the angle between y and the positive x -axis be β . Then, the angle between x and y is given by $\theta = \beta - \alpha$. We have:

$$x_1 = \|x\| \cos \alpha, \quad x_2 = \|x\| \sin \alpha,$$

$$y_1 = \|y\| \cos \beta, \quad y_2 = \|y\| \sin \beta.$$



Therefore, the inner product is:

$$\begin{aligned} \langle x, y \rangle &= x_1 y_1 + x_2 y_2 \\ &= \|x\| \cos \alpha \cdot \|y\| \cos \beta + \|x\| \sin \alpha \cdot \|y\| \sin \beta \\ &= \|x\| \|y\| (\cos \alpha \cos \beta + \sin \alpha \sin \beta) \\ &= \|x\| \|y\| \cos(\beta - \alpha) \\ &= \|x\| \|y\| \cos \theta. \end{aligned}$$

□

Remark

From the theorem, we can derive the formula for the angle θ between two vectors x and y :

$$\theta = \arccos \left(\frac{\langle x, y \rangle}{\|x\| \|y\|} \right).$$

Note that if $x \perp y$ (i.e., x is perpendicular to y), then $\langle x, y \rangle = 0$ and thus $\theta = \frac{\pi}{2}$.

Examples

$C(a, b)$ is the set of continuous functions on the interval (a, b) . For any $f, g \in C(a, b)$, we define the inner product as:

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx.$$

We can add a weighted inner product as follows:

$$\langle f, g \rangle_w = \int_a^b f(x)g(x)w(x) dx,$$

where $w(x)$ is a weight function such as $w(x) = e^{-x}$ or $w(x) = 1/x^2$.

We might as well use the family of **orthogonal functions**:

$$\{\cos nx, \sin nx \mid n = 0, 1, 2, \dots\} \quad \text{Fourier family.}$$

$$0 = \int_0^{2\pi} \cos nx \sin nx dx = \int_0^{2\pi} \cos nx \cos mx dx = \int_0^{2\pi} \sin nx \sin mx dx, \quad n \neq m.$$

1.8 Vector Product in $\mathbb{R}^3$

Let $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$ be two vectors in $\mathbb{R}^3$. The **vector product** (or cross product) of x and y is defined as:

$$x \times y = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = \begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} \mathbf{k},$$

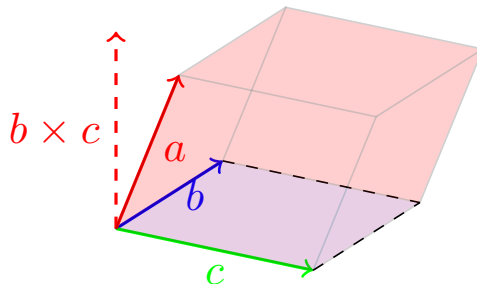
where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the standard unit vectors in $\mathbb{R}^3$.

1.8.1 Geometric Interpretation

To visualize the vector product, consider the triple product $a \cdot (b \times c)$, which represents the volume of the parallelepiped formed by the vectors a, b , and c :

$$a \cdot (b \times c) = (a_1, a_2, a_3) \cdot \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix} \mathbf{k} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

If $a \in \text{span}\{b, c\}$, then the volume is zero, indicating that the vectors are coplanar.



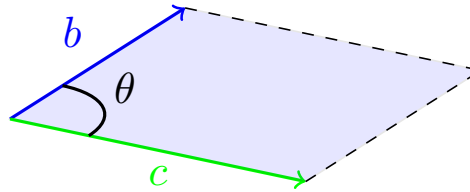
Also,

$$\begin{aligned}\|b \times c\|^2 &= \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix}^2 + \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix}^2 + \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}^2 = (b_2c_3 - b_3c_2)^2 + (b_1c_3 - b_3c_1)^2 + (b_1c_2 - b_2c_1)^2 = \\ &= (b_1^2 + b_2^2 + b_3^2)(c_1^2 + c_2^2 + c_3^2) - (b_1c_1 + b_2c_2 + b_3c_3)^2 = \|b\|^2\|c\|^2 - (b \cdot c)^2 = \|b\|^2\|c\|^2 \sin^2 \theta.\end{aligned}$$

Thus, we have:

$$\|b \times c\| = \|b\|\|c\| \sin \theta,$$

where θ is the angle between vectors b and c . This represents the area of the parallelogram formed by b and c .



2 Limits, Continuity, and Differentiability

2.1 Level Curves

Let $A \subset \mathbb{R}^n$ be a subset of $\mathbb{R}^n$ and let $f : A \rightarrow \mathbb{R}$ be a real-valued function defined on A . For any constant $c \in \mathbb{R}$, the **level set** of f corresponding to the value c is defined as:

$$L_c = \{x \in A \mid f(x) = c\}.$$

In the case where $n = 2$, the level sets are called **level curves** or **contour lines**. In the case where $n = 3$, the level sets are called **level surfaces**.

Example (Chapter 12, Ex 3)

Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = xy.$$

The level curves of f for different values of c are given by:

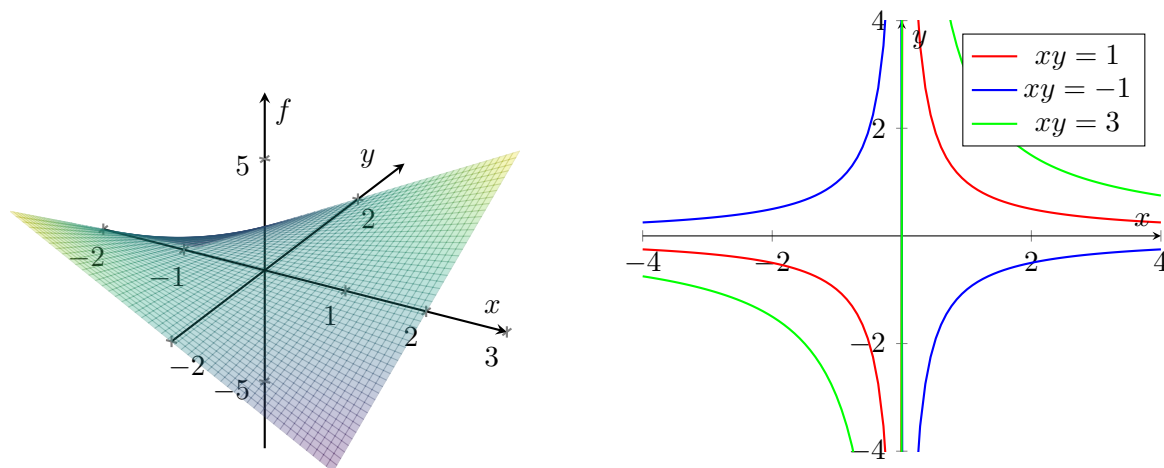
- For $c = -1$: $L_{-1} = \{(x, y) \in \mathbb{R}^2 \mid xy = -1\}$.
- For $c = 1$: $L_1 = \{(x, y) \in \mathbb{R}^2 \mid xy = 1\}$.
- For $c = 3$: $L_3 = \{(x, y) \in \mathbb{R}^2 \mid xy = 3\}$.

2.1.1 Graph of a Function

The **graph** of a function $f : A \rightarrow \mathbb{R}$ is defined as the set of points in $\mathbb{R}^{n+1}$ given by:

$$G_f = \{(x, f(x)) \mid x \in A\}.$$

If for instance, $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, then the graph of f is a surface in $\mathbb{R}^3$.



Remark

Level sets $F : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$; U open set:

$$F^{-1}(c) = \{x \in U \mid F(x) = c\}, \quad c = (c_1, c_2, \dots, c_m)$$

In terms of the coordinate functions $F = (f_1, f_2, \dots, f_m)$, we have:

$$F(x)(c), \text{ for } x \in \mathbb{R}^n \implies f_i(x_1, x_2, \dots, x_n) = c_i, \quad i = 1, 2, \dots, m.$$

Hence,

$$F^{-1}(c) = \bigcap_{i=1}^m \{x \in \mathbb{R}^n \mid f_i(x) = c_i\}.$$

Example

Let $F : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by:

$$F(x, y, z) = (x^2 + y^2 + z^2 - 1, 2x^2 + 2y^2 - 1).$$

The coordinate functions are:

$$f_1^{-1}(0) = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 - 1 = 0\},$$

$$f_2^{-1}(0) = \{(x, y, z) \in \mathbb{R}^3 \mid 2x^2 + 2y^2 - 1 = 0\}.$$

To find the level set we solve the system of equations:

$$x^2 + y^2 + z^2 - 1 = 0,$$

$$2x^2 + 2y^2 - 1 = 0.$$

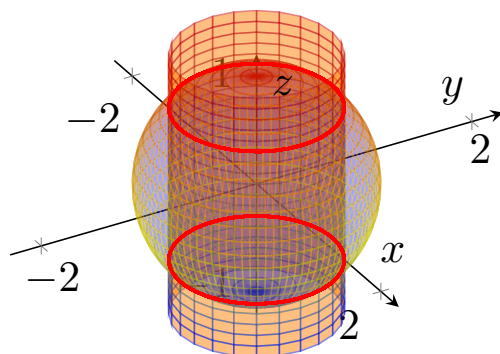
From the second equation, we have:

$$x^2 + y^2 = \frac{1}{2}.$$

Substituting this into the first equation gives:

$$\frac{1}{2} + z^2 - 1 = 0 \implies z^2 = \frac{1}{2} \implies z = \pm \frac{1}{\sqrt{2}}.$$

Thus, the level set $F^{-1}(0)$ is the intersection of the sphere $x^2 + y^2 + z^2 = 1$ and the cylinder $2x^2 + 2y^2 = 1$.



Remarks

- There is a connection between level sets and graphs. Firstly, the graph of a function $f : A \subset \mathbb{R}^n \rightarrow \mathbb{R}$ can be viewed as the level set of the function $F : A \times \mathbb{R} \rightarrow \mathbb{R}$ defined by:

$$F : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad F(x, z) = f(x) - z.$$

- The level curves allow us to draw tri-dimensional surfaces on a two-dimensional plane.
- The distance between two level curves (they can never intersect) gives us information about the steepness of the surface. If the level curves are close together, the surface is steep; if they are far apart, the surface is flat.

2.2 Limits of Functions

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a function, and let $a \in \mathbb{R}^n$ be a point in its domain. We say that the limit of $f(x)$ as x approaches a is $L \in \mathbb{R}^m$, denoted by:

$$\lim_{x \rightarrow a} f(x) = L,$$

if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x \in \mathbb{R}^n$ satisfying $0 < \|x - a\| < \delta$, we have:

$$\|f(x) - L\| < \epsilon.$$

2.2.1 Methods to Compute Limits

1. **Applying the Definition:** We must prove that for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$0 < (x, y) - (a, b) < \delta \implies |f(x, y) - L| < \epsilon.$$

2. **Iterative limits:**

$$\lim_{x \rightarrow a} \left(\lim_{y \rightarrow b} f(x, y) \right) \neq \lim_{y \rightarrow b} \left(\lim_{x \rightarrow a} f(x, y) \right).$$

This gives non-existence of the limit, so we cannot prove existence by this method.

3. **Following families of functions:** If we can find two different paths approaching the same point that yield different limit values, then the limit does not exist.

2.3 Higher Order Derivatives

Partial derivatives of first order $f(x_1, x_2, \dots, x_n)$

$$\frac{\partial f}{\partial x_1}, \quad \frac{\partial f}{\partial x_2}, \quad \dots, \quad \frac{\partial f}{\partial x_n}.$$

Second order partial derivatives:

$$\frac{\partial^2 f}{\partial x_1^2}, \quad \frac{\partial^2 f}{\partial x_i \partial x_j}, \quad \dots, \quad \frac{\partial^2 f}{\partial x_n^2}.$$

In general, the k -th order partial derivatives of f are denoted by:

$$\frac{\partial^k f}{\partial x_{i_1} \partial x_{i_2} \dots \partial x_{i_k}}, \quad \text{where } x_{i_j} \neq x_{i_k}.$$

Remark

We can say that the function f is of class C^k if all its partial derivatives up to order k exist and are continuous. Also, a function is **analytic** if it is of class C^∞ .

Example

Consider the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by:

$$f(x, y, z) = e^{xy} + z \cos x.$$

The first order partial derivatives are:

$$\frac{\partial f}{\partial x} = ye^{xy} - z \sin x, \quad \frac{\partial f}{\partial y} = xe^{xy}, \quad \frac{\partial f}{\partial z} = \cos x.$$

The second order partial derivatives are:

$$\frac{\partial^2 f}{\partial x^2} = y^2 e^{xy} - z \cos x, \quad \frac{\partial^2 f}{\partial y \partial x} = e^{xy} + xy e^{xy}, \quad \frac{\partial^2 f}{\partial y^2} = x^2 e^{xy}, \quad \frac{\partial^2 f}{\partial z^2} = 0 \dots$$

Remark

We call $\frac{\partial^k f}{\partial x_{i_1} \partial x_{i_2} \dots \partial x_{i_k}}$ a **mixed or crossing partial derivative** if at least two of the indices i_j are different. For example, $\frac{\partial^2 f}{\partial x \partial y}$ is a mixed partial derivative, while $\frac{\partial^2 f}{\partial x^2}$ is not.

2.3.1 Schwarz's Theorem

Take a function $f : A \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ of class C^1 . Then, the mixed partial derivatives of f are equal:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

Remark

In general, if $f : A \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is of class C^{k-1} , then the order of differentiation does not matter for any mixed partial derivatives of order less than or equal to k . That is, for any indices $i_1, i_2, \dots, i_k$ and any permutation σ of $\{1, 2, \dots, k\}$, we have:

$$\frac{\partial^k f}{\partial x_{i_1} \partial x_{i_2} \dots \partial x_{i_k}} = \frac{\partial^k f}{\partial x_{i_{\sigma(1)}} \partial x_{i_{\sigma(2)}} \dots \partial x_{i_{\sigma(k)}}}.$$

Proof. To simplify the proof, we assume that the derivatives are taken at the point $(0, 0)$. Take an arbitrary point (x, y) inside a ball centered at $(0, 0)$:

$$\|(x, y)\| < \varepsilon, \quad \text{assuming } B_\varepsilon(0, 0).$$

Then, we can **discretize** the derivatives as follows:

$$\begin{aligned} \frac{\partial f(x, y)}{\partial x} &\approx \frac{f(x+h, y) - f(x-h, y)}{2h}, \\ \frac{\partial f(x, y)}{\partial y} &\approx \frac{f(x, y+k) - f(x, y-k)}{2k}. \end{aligned}$$

And for the second order derivatives:

$$\begin{aligned} \frac{\partial^2 f(x, y)}{\partial x^2} &\approx \frac{f(x+h, y) - 2f(x, y) + f(x-h, y)}{h^2}, \\ \frac{\partial^2 f(x, y)}{\partial y^2} &\approx \frac{f(x, y+k) - 2f(x, y) + f(x, y-k)}{k^2}. \end{aligned}$$

For the mixed partial derivative, we have:

$$\frac{\partial^2 f(x, y)}{\partial x \partial y} \approx \frac{f(x+h, y+k) - f(x+h, y-k) - f(x-h, y+k) + f(x-h, y-k)}{4hk}.$$

Back to the proof, we use the second differentiation:

$$D(x, y) = f(x, y) - f(x, 0) - f(0, y) + f(0, 0).$$

Now, we define the function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ by:

$$\left. \begin{aligned} \varphi(x) &= f(x, y) - f(x, 0), \\ \varphi(0) &= f(0, y) - f(0, 0). \end{aligned} \right\} \implies D(x, y) = \varphi(x) - \varphi(0).$$

Then, we have:

$$\varphi'(x) = \frac{\partial f(x, y)}{\partial x} - \frac{\partial f(x, 0)}{\partial x},$$

and by the Mean Value Theorem, there exists $-x \leq \hat{x} \leq x$ such that:

$$D(x, y) = \varphi(x) - \varphi(0) = x\varphi'(\hat{x}) = x \left(\frac{\partial f(\hat{x}, y)}{\partial x} - \frac{\partial f(\hat{x}, 0)}{\partial x} \right).$$

Applying the Mean Value Theorem again, now for the map:

$$y \mapsto \frac{\partial f(\hat{x}, y)}{\partial x},$$

we find that

$$D(x, y) = xy \frac{\partial^2 f(\hat{x}, \hat{y})}{\partial y \partial x}, \quad \text{for some } -y \leq \hat{y} \leq y$$

Similarly, we define the function $\psi : \mathbb{R} \rightarrow \mathbb{R}$ by:

$$\left. \begin{aligned} \psi(y) &= f(x, y) - f(0, y), \\ \psi(0) &= f(x, 0) - f(0, 0). \end{aligned} \right\} \implies D(x, y) = \psi(y) - \psi(0).$$

By following the same steps, we find that:

$$D(x, y) = xy \frac{\partial^2 f(\tilde{x}, \tilde{y})}{\partial x \partial y}, \quad \text{for some } -x \leq \tilde{x} \leq x, -y \leq \tilde{y} \leq y.$$

Since f is of class C^1 , we have:

$$\frac{\partial^2 f(\hat{x}, \hat{y})}{\partial y \partial x} = \frac{\partial^2 f(\tilde{x}, \tilde{y})}{\partial x \partial y}.$$

Passing to the limit as $(x, y) \rightarrow (0, 0)$, we conclude that:

$$\frac{\partial^2 f(0, 0)}{\partial y \partial x} = \frac{\partial^2 f(0, 0)}{\partial x \partial y}.$$

□

3 Local Extrema

Let $f : A \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a function, and let $a \in A$ be a point in its domain. We say that f has a **local maximum** at a if there exists a neighborhood U of a such that for all $x \in U$, we have:

$$f(x) \leq f(a).$$

Similarly, f has a **local minimum** at a if there exists a neighborhood U of a such that for all $x \in U$, we have:

$$f(x) \geq f(a).$$

Definition

Given a function $f : A \subset \mathbb{R}^n \rightarrow \mathbb{R}$ and a point $a \in A$, we say that a is a **critical point** or **stationary point** of f if the gradient of f at a is zero, i.e.,

$$\nabla f(a) = 0.$$

Remark

A critical point that is not a local extremum is called a **saddle point**.

- **Theorem:** If $f : A \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable and has a local extremum at a , then a is a critical point of f .
- If a is a local extremum, and the directional derivative of f at a in any direction v exists, then the directional derivative must be zero:

$$D_v f(a) = \nabla f(a) \cdot v = 0.$$

- If f is differentiable, and we find that $\nabla f(a) = 0$, then every directional derivative at a is zero.
- If any of the directional derivatives at a does not exist, then the function is not differentiable at a , and a is an **angular point**.
- In both cases we can find a potential local extremum at a , but it can also be a saddle point.

3.1 How to find local extrema?

Let's consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = 3x^2 + y^2 - 6x - 4y + 8.$$

To find the local extrema of f , we first compute the gradient:

$$\nabla f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (6x - 6, 2y - 4).$$

Setting the gradient equal to zero gives us the critical point:

$$6x - 6 = 0 \implies x = 1, \quad 2y - 4 = 0 \implies y = 2.$$

Thus, the critical point is $(1, 2)$. We have:

$$f(1, 2) = 3(1)^2 + (2)^2 - 6(1) - 4(2) + 8 = 3 + 4 - 6 - 8 + 8 = 1.$$

We will complete the squares to determine the nature of the critical point:

$$\begin{aligned} f(x, y) &= 3x^2 - 6x + y^2 - 4y + 8 \\ &= 3(x^2 - 2x) + (y^2 - 4y) + 8 \\ &= 3(x - 1)^2 - 3 + (y - 2)^2 - 4 + 8 \\ &= 3(x - 1)^2 + (y - 2)^2 + 1. \end{aligned}$$

Since $f(x, y) \geq 1 = f(1, 2)$ for all $(x, y) \in \mathbb{R}^2$, the critical point $(1, 2)$ is a local minimum of f .

Example

Find the extrema of the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = 1 - \sqrt{x^2 + y^2}.$$

We are looking for the points where:

- The partial derivatives are zero (stationary points), or
- The function is not differentiable (angular points).

First, we compute the partial derivatives:

$$\frac{\partial f}{\partial x} = -\frac{x}{\sqrt{x^2 + y^2}}, \quad \frac{\partial f}{\partial y} = -\frac{y}{\sqrt{x^2 + y^2}}.$$

Setting the partial derivatives equal to zero gives us:

$$-\frac{x}{\sqrt{x^2 + y^2}} = 0 \implies x = 0, \quad -\frac{y}{\sqrt{x^2 + y^2}} = 0 \implies y = 0.$$

Thus, the function f is not differentiable at $(0, 0)$ since the partial derivatives are undefined at that point. However, we can still evaluate f at $(0, 0)$:

$$f(0, 0) = 1 - \sqrt{0^2 + 0^2} = 1.$$

For any other point $(x, y) \neq (0, 0)$, we have:

$$f(x, y) = 1 - \sqrt{x^2 + y^2} < 1.$$

Thus, the point $(0, 0)$ is a global maximum of f .

Example

Another example is the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = 1 - x^2 + y^2.$$

We compute the partial derivatives:

$$\left. \begin{array}{l} \frac{\partial f}{\partial x} = -2x \\ \frac{\partial f}{\partial y} = 2y \end{array} \right\} \implies (0, 0) \text{ is a stationary point.}$$

Evaluating f at $(0, 0)$ gives us:

$$f(0, 0) = 1 - 0^2 - 0^2 = 1.$$

For any other point $(x, y) \neq (0, 0)$, we have:

$$f|_{x=0}(x, y) = f(0, y) = 1 + y^2 > 1, \quad f|_{y=0}(x, y) = f(x, 0) = 1 - x^2 < 1.$$

Thus, the point $(0, 0)$ is a saddle point of f .

3.2 Methods for Classification of Extrema

To classify the extrema systematically, we will use:

- Quadratic forms
- The Hessian matrix.

3.2.1 Second Derivative (general definition)

Take a vector $v = (v_1, \dots, v_n) \in \mathbb{R}^n$ and a function $f : A \subset \mathbb{R}^n \rightarrow \mathbb{R}$, differentiable at a . Then,

$$D_v f(a) = \langle \nabla f(a), v \rangle = \sum_{i=1}^n v_i \cdot \frac{\partial f}{\partial x_i}(a).$$

Then, we can define the second derivative of f at a in the direction of v as:

$$D_v^2 f(a) = D_v(D_v f)(a) = \sum_{j=1}^n v_j \cdot \frac{\partial}{\partial x_j} \left(\sum_{i=1}^n v_i \cdot \frac{\partial f}{\partial x_i}(a) \right) = \sum_{i=1}^n \sum_{j=1}^n v_i v_j \cdot \frac{\partial^2 f}{\partial x_i \partial x_j}(a) = v^T H_f(a) v,$$

where $H_f(a)$ is the Hessian matrix of f at a .

$$\boxed{D_v^2 f(a) = v^T H_f(a) v}$$

3.2.2 Hessian Matrix

The **Hessian matrix** of a function $f : A \subset \mathbb{R}^n \rightarrow \mathbb{R}$ at a point $a \in A$ is the $n \times n$ matrix given by:

$$H_f(a) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(a) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(a) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(a) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(a) & \frac{\partial^2 f}{\partial x_2^2}(a) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(a) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(a) & \frac{\partial^2 f}{\partial x_n \partial x_2}(a) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(a) \end{bmatrix}.$$

Example

Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = \log(x^2 + y^3)$$

The partial derivatives are:

$$\frac{\partial f}{\partial x} = \frac{2x}{x^2 + y^3}, \quad \frac{\partial f}{\partial y} = \frac{3y^2}{x^2 + y^3}.$$

The second order partial derivatives are:

$$\frac{\partial^2 f}{\partial x^2} = \frac{2(y^3 - x^2)}{(x^2 + y^3)^2}, \quad \frac{\partial^2 f}{\partial y^2} = \frac{6yx^2 - 3y^4}{(x^2 + y^3)^2}, \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{-6xy^2}{(x^2 + y^3)^2}.$$

Thus, the Hessian matrix of f at (x, y) is:

$$H_f(x, y) = \begin{bmatrix} \frac{2(y^3 - x^2)}{(x^2 + y^3)^2} & \frac{-6xy^2}{(x^2 + y^3)^2} \\ \frac{-6xy^2}{(x^2 + y^3)^2} & \frac{6yx^2 - 3y^4}{(x^2 + y^3)^2} \end{bmatrix}.$$

3.2.3 Quadratic Forms

A **quadratic form** on $\mathbb{R}^n$ is a function $Q : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by:

$$Q(x) = \sum_{i,j=1}^n a_{ij}x_i x_j = X^T A X = a_{11}x_1^2 + a_{22}x_2^2 + \cdots + a_{nn}x_n^2 + \sum_{\substack{1 \leq i,j \leq n \\ i \neq j}} a_{ij}x_i x_j,$$

where a_{ij} are constants. For example, the function $Q_1 : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$Q_1(x_1, x_2) = x_1^2 - x_2^2$$

is a quadratic form on $\mathbb{R}^2$, and so is the function $Q_2 : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by:

$$Q_2(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 5x_3^2 + 2x_1x_2 - 2x_1x_3 + 2x_2x_3.$$

Quadratic forms can be classified as follows:

- A quadratic form Q is **positive definite** if $Q(x) > 0$ for all $x \in \mathbb{R}^n$.
- A quadratic form Q is **negative definite** if $Q(x) < 0$ for all $x \in \mathbb{R}^n$.
- A quadratic form Q is **indefinite** if $\exists x, y \in \mathbb{R}^n$ such that $Q(x) > 0$ and $Q(y) < 0$.

Any quadratic form Q can be associated with a symmetric matrix A such that $Q(x) = x^T A x$. For example, the quadratic form $Q_2(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 5x_3^2 + 2x_1x_2 - 2x_1x_3 + 2x_2x_3$ can be represented by the symmetric matrix:

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 5 \end{bmatrix}.$$

Examples

Consider the quadratic form $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by:

$$\|x\|^2 = x^T I x = \sum_{i=1}^n x_i^2,$$

where I is the identity matrix. This is a positive definite quadratic form.

The quadratic form $Q_1 : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$Q(x_1, x_2) = x_1^2 - x_2^2$$

is an indefinite quadratic form, since $Q(1, 0) = 1 > 0$ and $Q(0, 1) = -1 < 0$.

The quadratic form $Q_2 : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by:

$$Q_2(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 5x_3^2 + 2x_1x_2 - 2x_1x_3 + 2x_2x_3$$

is a semi-positive definite quadratic form, since it can be rewritten as:

$$Q_2(x_1, x_2, x_3) = (x_1 + x_2 - x_3)^2 + (x_2 + 2x_3)^2,$$

which is a sum of squares and hence always non-negative.

Proposition

A quadratic form $Q : \mathbb{R}^n \rightarrow \mathbb{R}$ can be expressed as a sum of squares of linear forms. That is, there exist linear forms $L_1, L_2, \dots, L_m : \mathbb{R}^n \rightarrow \mathbb{R}$ such that:

$$Q(x_1, x_2, \dots, x_n) = L_1^2 \pm L_2^2 \pm \cdots \pm L_n^2,$$

where $L_i, \dots, L_n$ do not depend on the variables $x_1, x_2, \dots, x_{i-1}$ for $i = 1, 2, \dots, n$.

Example

Consider the quadratic form $Q : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by:

$$Q(x_1, x_2, x_3) = x_1^2 - x_2^2 + x_3^2 + 2x_1x_2 - 4x_1x_3.$$

We can rewrite this as:

$$\begin{aligned} Q(x_1, x_2, x_3) &= (x_1 + x_2 - 2x_3)^2 - 2x_2^2 - 2x_3^2 \\ &= (x_1 + x_2 - 2x_3)^2 - 2x_2^2 + 4x_2x_3 - 3x_3^2 \\ &= (x_1 + x_2 - 2x_3)^2 - 2(x_2 - x_3)^2 - x_3^2. \end{aligned}$$

Remark

If we do not have squares in the expression of a quadratic form, we can always complete the squares to rewrite it as a sum of squares of linear forms. For example, the quadratic form $Q : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$\begin{aligned} Q(x_1, x_2) &= x_1x_2 + 2x_2x_3 \begin{cases} x_1 = y_1 - y_2, \\ x_2 = y_1 + y_2 \\ x_3 = y_3 \end{cases} \implies \\ \implies Q(y_1, y_2, y_3) &= y_1^2 - y_2^2 + 2y_1y_3 + 2y_2y_3 \end{aligned}$$

Remark

Given X^TAX , we can express $A = PDP^{-1}$ (A symmetric) and with all the eigenvalues of A in D being real,

$$A = PDP^{-1}, \quad D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

where λ_i are the eigenvalues of A . If we now perform the change of variables $X = PY$, we have:

$$Q(X) = X^TAX = (PY)^T A(PY) = Y^T P^T A P Y = Y^T D Y = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2.$$

Theorem

Let $Q : \mathbb{R}^n \rightarrow \mathbb{R}$ be a quadratic form. Then:

- Q is positive definite if and only if all the eigenvalues of the associated matrix A are positive.
- Q is negative definite if and only if all the eigenvalues of the associated matrix A are negative.
- Q is indefinite if and only if there exist both positive and negative eigenvalues of the associated matrix A .

3.2.4 Taylor's Polynomial

Let $f : A \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be a differentiable function, and let $a = (x_0, y_0) \in A$. The **Taylor's polynomial** of f at a is the polynomial defined by:

$$f(x, y) = f(x_0, y_0) + \nabla f(x_0, y_0) \cdot \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} + \frac{1}{2}(x - x_0, y - y_0) \cdot H_f(x_0, y_0) \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} + R,$$

where $H_f(x_0, y_0)$ is the Hessian matrix of f at a .

One can see that it is composed of three terms:

- The first term $f(x_0, y_0)$ is the value of the function at the point a .
- The second term $\nabla f(x_0, y_0) \cdot \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}$ is the linear approximation of f at a .
- The third term $\frac{1}{2}(x - x_0, y - y_0) \cdot H_f(x_0, y_0) \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}$ is the quadratic approximation of f at a .

3.3 Classification of Extrema

Let $U \subset \mathbb{R}^n$ be an open set, and let $f : U \rightarrow \mathbb{R}$ be a function of class C^1 . Let $a \in U$ be a critical point of f . If

$$\nabla f(a) = \left(\frac{\partial f}{\partial x_1}(a), \frac{\partial f}{\partial x_2}(a), \dots, \frac{\partial f}{\partial x_n}(a) \right) = 0,$$

then

$$f(x) - f(a) = \frac{1}{2}(x - a)^T H_f(a)(x - a),$$

where $H_f(a)$ is the Hessian matrix of f at a . Thus, the nature of the critical point a can be determined by the definiteness of the quadratic form associated with the Hessian matrix:

- If the quadratic form $Q(x) = (x - a)^T H_f(a)(x - a)$ is positive definite, then a is a local minimum of f .
- If the quadratic form $Q(x) = (x - a)^T H_f(a)(x - a)$ is negative definite, then a is a local maximum of f .
- If the quadratic form $Q(x) = (x - a)^T H_f(a)(x - a)$ is indefinite, then a is a saddle point of f .

3.3.1 Hessian or Sylvester's Criteria in $\mathbb{R}^2$

Given the Hessian matrix of a function $f : A \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ at a critical point $a = (x, y)$:

$$H_f(a) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2}(a) & \frac{\partial^2 f}{\partial x \partial y}(a) \\ \frac{\partial^2 f}{\partial y \partial x}(a) & \frac{\partial^2 f}{\partial y^2}(a) \end{bmatrix},$$

the determinant of the Hessian matrix is given by:

$$\det(H_f(a)) = \frac{\partial^2 f}{\partial x^2}(a) \cdot \frac{\partial^2 f}{\partial y^2}(a) - \left(\frac{\partial^2 f}{\partial x \partial y}(a) \right)^2.$$

Then, the classification of the critical point $a = (x_0, y_0)$ can be determined as follows:

- If $\det(H_f(x_0, y_0)) > 0$ and $\frac{\partial^2 f}{\partial x^2}(x_0, y_0) > 0$, then a is a local minimum of f .
- If $\det(H_f(x_0, y_0)) > 0$ and $\frac{\partial^2 f}{\partial x^2}(x_0, y_0) < 0$, then a is a local maximum of f .
- If $\det(H_f(x_0, y_0)) < 0$, then a is a saddle point of f .
- If $\det(H_f(x_0, y_0)) = 0$, then the test is inconclusive, and we need to use higher-order derivatives or other methods to classify the critical point.

Example

Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = x^2 + 2y^2 - 4y.$$

The gradient of f is:

$$\nabla f(x, y) = (2x, 4y - 4).$$

Setting the gradient equal to zero gives us the critical point:

$$2x = 0 \implies x = 0, \quad 4y - 4 = 0 \implies y = 1.$$

Thus, the critical point is $(0, 1)$. The Hessian matrix of f at $(0, 1)$ is:

$$H_f(0, 1) = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}.$$

The determinant of the Hessian matrix is:

$$\det(H_f(0, 1)) = (2)(4) - (0)^2 = 8 > 0,$$

and $\frac{\partial^2 f}{\partial x^2}(0, 1) = 2 > 0$. Therefore, the critical point $(0, 1)$ is a local minimum of f .

Example

Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = 8x^3 - 24xy + 3y^2.$$

The gradient of f is:

$$\nabla f(x, y) = (24x^2 - 24y, -24x + 3y^2).$$

Setting the gradient equal to zero gives us the critical points:

$$\begin{aligned} x^2 - y = 0 &\implies y = x^2, \\ -8x + y^2 = 0 &\implies y^2 = 8x. \end{aligned}$$

Substituting $y = x^2$ into $y^2 = 8x$ gives us:

$$x^4 = 8x \implies x^4 - 8x = 0 \implies x(x^3 - 8) = 0 \implies x = 0 \text{ or } x^3 = 8 \implies x = 2.$$

Thus, the critical points are $(0, 0)$ and $(2, 4)$. The Hessian matrices of f at $(0, 0)$ and $(2, 4)$ are:

$$H_f(0, 0) = \begin{bmatrix} 0 & -24 \\ -24 & 0 \end{bmatrix}, \quad H_f(2, 4) = \begin{bmatrix} 96 & -24 \\ -24 & 24 \end{bmatrix}.$$

The determinants of the Hessian matrix at said points are:

$$\begin{aligned}\det(H_f(0,0)) &= (0)(0) - (-24)^2 = -576 < 0, \\ \det(H_f(2,4)) &= (96)(24) - (-24)^2 = 2304 - 576 = 1728 > 0.\end{aligned}$$

Since $\det(H_f(0,0)) < 0$, the critical point $(0,0)$ is a saddle point of f . Since $\det(H_f(2,4)) > 0$ and $\frac{\partial^2 f}{\partial x^2}(2,4) = 96 > 0$, the critical point $(2,4)$ is a local minimum of f .

3.3.2 Extension to $\mathbb{R}^3$

The hessian matrix of a function $f : A \subset \mathbb{R}^3 \rightarrow \mathbb{R}$ is of the form:

$$H_f = \begin{bmatrix} A & B & C \\ D & E & F \\ G & H & I \end{bmatrix} \approx \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}.$$

Looking at the determinants of the leading principal minors of H_f , we have:

- $\Delta_1 = A$
- $\Delta_2 = \det \begin{pmatrix} A & B \\ D & E \end{pmatrix} = AE - BD$
- $\Delta_3 = \det(H_f) = A(EI - FH) - B(DI - FG) + C(DH - EG)$

With the eigenvalues of H_f being $\lambda_1, \lambda_2, \lambda_3$, we have:

- If $\Delta_1 > 0$, $\Delta_2 > 0$, and $\Delta_3 > 0$, then f is positive definite, and the critical point is a local minimum.
- If $\Delta_1 < 0$, $\Delta_2 > 0$, and $\Delta_3 < 0$, then f is negative definite, and the critical point is a local maximum.
- Any other combination of signs for Δ_1, Δ_2 , and Δ_3 indicates that the critical point is a saddle point.

4 Implicit Function Theorem

Using systems of linear equations as a base, we discuss the implications of the **Implicit Function Theorem** for level sets. In other words, the role of differentiability in the local structure of level sets. To do so, we use the linear approximation of a differentiable function to see that level sets are locally graphs.

Let U be an open set, and $F = (f_1, f_2, \dots, f_m) : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a differentiable function at any point in U . The level set

$$F^{-1}(c) = \bigcap_{i=1}^m f_i^{-1}(c_i),$$

where $c = (c_1, c_2, \dots, c_m) \in \mathbb{R}^m$, is the set of points in U that satisfy the system of equations:

$$F(x_1, \dots, x_n) = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n)) = (c_1, \dots, c_m),$$

or equivalently,

$$\begin{cases} f_1(x_1, \dots, x_n) = c_1 \\ f_2(x_1, \dots, x_n) = c_2 \\ \vdots \\ f_m(x_1, \dots, x_n) = c_m \end{cases}.$$

Remark

If f_i are linear functions, then we have a system of linear equations, and we know different procedures to find independent variables. The Implicit Function Theorem is a generalization of this idea to the case where f_i are differentiable functions, based on the linear approximation of f_i at a point in the level set. Indeed, fix a point $a = (a_1, \dots, a_n) \in F^{-1}(c)$, and $\mathbf{x} \in \mathbb{R}^n$ very close to 0. Then, we have:

$$a + \mathbf{x} \in F^{-1}(c) \quad F(a + \mathbf{x}) = c$$

$$F(a + \mathbf{x}) = F(a) + F'(a) \cdot \mathbf{x} + G(\mathbf{x}),$$

where $G : U \rightarrow M_{M \times N}$ is a function such that $G(\mathbf{x}) \rightarrow 0$ as $\mathbf{x} \rightarrow 0$, and consequently,

$$F'(a) \cdot \mathbf{x} \rightarrow 0 \text{ as } \mathbf{x} \rightarrow 0.$$

Assume $N \geq M$ and we find that

$$\left. \begin{array}{l} \frac{\partial f_1(a)}{\partial x_1} x_1 + \frac{\partial f_1(a)}{\partial x_2} x_2 + \dots + \frac{\partial f_1(a)}{\partial x_N} x_N = 0 \\ \frac{\partial f_2(a)}{\partial x_1} x_1 + \frac{\partial f_2(a)}{\partial x_2} x_2 + \dots + \frac{\partial f_2(a)}{\partial x_N} x_N = 0 \\ \vdots \\ \frac{\partial f_m(a)}{\partial x_1} x_1 + \frac{\partial f_m(a)}{\partial x_2} x_2 + \dots + \frac{\partial f_m(a)}{\partial x_N} x_N = 0 \end{array} \right\} \implies JF(a) \cdot \mathbf{x} = 0.$$

Thanks to the Algebra we have:

$$\begin{aligned}
\text{rank}(F'(a)) = M &\iff \text{The } M \text{ rows of } F'(a) \text{ are linearly independent} \\
&\iff \text{There exist } M \text{ columns of } F'(a) \text{ that are linearly independent} \\
&\iff F'(a) \text{ contains } M \text{ columns such that the } M \times M \text{ submatrix formed} \\
&\quad \text{by these columns has non-zero determinant.} \\
&\iff \text{The space of solutions to that linear system has dimension } N - M. \\
&\iff F \text{ has maximal rank at } a. \\
&\iff \{\nabla f_1(a), \nabla f_2(a), \dots, \nabla f_m(a)\} \text{ are linearly independent vectors in } \mathbb{R}^n.
\end{aligned}$$

4.1 Manifolds

Let $1 \leq m \leq n - 1$ and $D \subset \mathbb{R}^n$ be a set we want to present as an **m-dimensional manifold**. Let $k = n - m$, and suppose a neighborhood of any point $a \in D$. Then, there exist $\mathcal{C}^1$ functions such that

- Near a , D can be described as the set of points that satisfy a system of k equations:

$$\begin{cases} f_1(x_1, \dots, x_n) = 0 \\ f_2(x_1, \dots, x_n) = 0 \\ \vdots \\ f_k(x_1, \dots, x_n) = 0 \end{cases}$$

- The gradients of the functions $f_1, f_2, \dots, f_k$ are linearly independent at a :

$$\nabla f_1(a), \nabla f_2(a), \dots, \nabla f_k(a) \text{ are linearly independent vectors in } \mathbb{R}^n.$$

4.2 Tangent Space

Let $F = (f_1, f_2, \dots, f_n)$ be a function, and consider the set of vectors $v \in \mathbb{R}^n$ such that $F'(a) \cdot v = 0$. Then,

$$\left. \begin{aligned} \frac{\partial f_1(a)}{\partial x_1} v_1 + \frac{\partial f_1(a)}{\partial x_2} v_2 + \dots + \frac{\partial f_1(a)}{\partial x_n} v_n &= 0 \\ \frac{\partial f_2(a)}{\partial x_1} v_1 + \frac{\partial f_2(a)}{\partial x_2} v_2 + \dots + \frac{\partial f_2(a)}{\partial x_n} v_n &= 0 \\ \vdots & \\ \frac{\partial f_m(a)}{\partial x_1} v_1 + \frac{\partial f_m(a)}{\partial x_2} v_2 + \dots + \frac{\partial f_m(a)}{\partial x_n} v_n &= 0 \end{aligned} \right\} \implies v \in T_a D,$$

where $T_a D$ is the **tangent space** to the manifold D at the point a . The tangent space is represented as:

$$a + \left\{ \sum_{i=1}^n \lambda_i \nabla f_i(a) : \lambda_i \in \mathbb{R} \right\}.$$

The set in the expression above is the span of the vectors $\nabla f_1(a), \nabla f_2(a), \dots, \nabla f_m(a)$, which are linearly independent.

Example

Consider $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + 2y^2 - 5z^2 = 1\}$. We will find the tangent space and normal line to S at the point $a = (2, -1, 1)$:

$$\text{With } S = f^{-1}(1), \quad f(x, y, z) = x^2 + 2y^2 - 5z^2,$$

we have:

$$\nabla f(x, y, z) = (2x, 4y, -10z).$$

At the point $a = (2, -1, 1)$, we have:

$$\nabla f(2, -1, 1) = (4, -4, -10).$$

Thus, the tangent space to S at the point a is given by:

$$\begin{aligned} T_a S &= \{(x, y, z) \in \mathbb{R}^3 : \nabla f(2, -1, 1) \cdot \begin{pmatrix} x-2 \\ y+1 \\ z-1 \end{pmatrix} = 0\} \\ &= \{(x, y, z) \in \mathbb{R}^3 : 4(x-2) - 4(y+1) - 10(z-1) = 0\} \\ &= \{(x, y, z) \in \mathbb{R}^3 : 2x - 2y - 5z = 1\}. \end{aligned}$$

and the normal line to S at the point a is given by:

$$\begin{aligned} L &= \{(x, y, z) \in \mathbb{R}^3 : (x, y, z) = (2, -1, 1) + t(4, -4, -10), t \in \mathbb{R}\} \\ &= \{(x, y, z) \in \mathbb{R}^3 : \frac{x-2}{4} = \frac{y+1}{-4} = \frac{z-1}{-10} = \lambda\} \\ &= \{(x, y, z) \in \mathbb{R}^3 : x = \frac{1}{2} + 2y + 5z, y = \lambda, z = \mu, \lambda, \mu \in \mathbb{R}\} \\ &= \{(x, y, z) \in \mathbb{R}^3 : \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1/2 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 5/2 \\ 0 \\ 1 \end{pmatrix}, \lambda, \mu \in \mathbb{R}\}. \end{aligned}$$

To construct the tangent plane to S at the point a , we can use the fact that the normal vector to the plane is given by $\nabla f(2, -1, 1) = (4, -4, -10)$. Thus,

$$\left. \begin{aligned} (4, -4, -10) \cdot (4, 4, 0) &= 0 \\ (4, -4, -10) \cdot (10, 0, 4) &= 0 \end{aligned} \right\} \implies \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 4 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 10 \\ 0 \\ 4 \end{pmatrix} \in T_a S, \quad \lambda, \mu \in \mathbb{R}.$$

5 Lagrange Multipliers

We would like to optimize a function $g : A \subset \mathbb{R}^n \rightarrow \mathbb{R}$ subject to the constraint

$$\begin{cases} f_1(x_1, \dots, x_n) = c_1 \\ f_2(x_1, \dots, x_n) = c_2 \\ \vdots \\ f_m(x_1, \dots, x_n) = c_m \end{cases},$$

where $F = (f_1, f_2, \dots, f_m) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a function. Then, we look for the critical points of g restricted to the level set

$$F^{-1}(c) = \bigcap_{i=1}^m f_i^{-1}(c_i).$$

F' has full rank so that the level set $F^{-1}(c)$ is a manifold, i.e., the gradients $\nabla f_i(a)$ are linearly independent at any point $a \in F^{-1}(c)$.

5.1 Necessary Condition for Optimality

Suppose g has a local maximum at $a \in F^{-1}(c)$. Then, using the Implicit Function Theorem, the set $F^{-1}(c) = \text{graph } \phi$

$$P = (P_1, \phi(P_1)) \in F^{-1}(c), \quad P_1 = (x_1, \dots, x_{n-m}) \in \mathbb{R}^{n-m},$$

where $\phi : \mathbb{R}^{n-m} \rightarrow \mathbb{R}^m$. Then, the function g restricted to the level set $F^{-1}(c)$ can be expressed as $g(P_1, \phi(P_1))$ and we have:

$$F^{-1}(c) = \text{graph}(\phi) \implies F(\mathbf{x}, \phi(\mathbf{x})) = c \text{ and therefore } f_i(\mathbf{x}, \phi(\mathbf{x})) = c_i \text{ for } i = 1, 2, \dots, m.$$

Now, what is the tangent space to the level set $F^{-1}(c)$? It is given by:

$$S = \{\nabla f_i(\mathbf{x}_1, \phi(\mathbf{x}_1)) \cdot \Delta(\mathbf{x}_1) = 0 \text{ for } i = 1, 2, \dots, m\},$$

where Δ is the Jacobian matrix of the function $\mathbf{x}_1 \mapsto (\mathbf{x}_1, \phi(\mathbf{x}_1))$ and $\mathbf{x}_1 = (x_1, \dots, x_{n-m})$. Then,

$$\phi_j(\mathbf{x}_1) = \phi_j(x_1, \dots, x_{n-m}), \quad j = 1, 2, \dots, m,$$

and consequently,

$$\Delta(\mathbf{x}_1) = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}, \quad \phi'(\mathbf{x}_1) = \begin{pmatrix} \frac{\partial \phi_1}{\partial x_1}(\mathbf{x}_1) & \frac{\partial \phi_1}{\partial x_2}(\mathbf{x}_1) & \cdots & \frac{\partial \phi_1}{\partial x_{n-m}}(\mathbf{x}_1) \\ \frac{\partial \phi_2}{\partial x_1}(\mathbf{x}_1) & \frac{\partial \phi_2}{\partial x_2}(\mathbf{x}_1) & \cdots & \frac{\partial \phi_2}{\partial x_{n-m}}(\mathbf{x}_1) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \phi_m}{\partial x_1}(\mathbf{x}_1) & \frac{\partial \phi_m}{\partial x_2}(\mathbf{x}_1) & \cdots & \frac{\partial \phi_m}{\partial x_{n-m}}(\mathbf{x}_1) \end{pmatrix}.$$

We know that $\nabla f_1, \dots, \nabla f_m$ are linearly independent on S (they form a basis). Finally, using the function we want to maximize (on the constraint),

$$\mathbf{x}_1 \mapsto g(\mathbf{x}_1, \phi(\mathbf{x}_1))$$

and differentiating g :

$$g(\mathbf{x}_1, \phi(\mathbf{x}_1)) \implies \nabla g|_{F^{-1}(c)} = \nabla g(\mathbf{x}_1, \phi(\mathbf{x}_1)) \cdot \Delta(\mathbf{x}_1) = 0, \text{ tangent plane to } g.$$

Hence, $\nabla g(\mathbf{x}_1, \phi(\mathbf{x}_1))$ belongs to S :

$$\nabla g(\mathbf{x}_1, \phi(\mathbf{x}_1)) \in \left\{ \sum_{i=1}^m \lambda_i \nabla f_i \right\}, \text{ such that } F(\mathbf{x}) = c.$$

Example: Chapter 2.3, Exercise 5

Compute the absolute maximum and minimum of the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by:

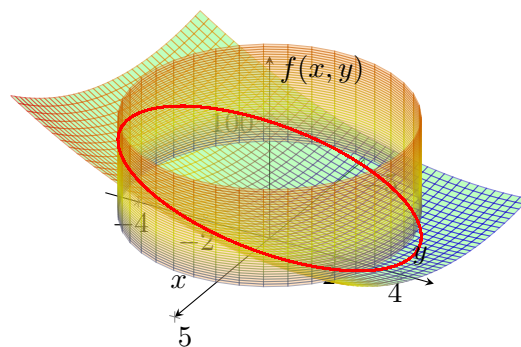
$$f(x, y, z) = x^2 + y^2 + 6x - 8y + 25$$

on the set $D = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 \leq 16\}$.

By completing the squares, we can rewrite f as:

$$f(x, y, z) = (x + 3)^2 + (y - 4)^2.$$

The function f is a paraboloid that opens upwards, and the set D is a cylinder of radius 4 centered at the origin.



The Weierstrass Theorem states that given a continuous function defined on a compact set, it attains its maximum and minimum values.

- The function f is continuous since it is a composition of continuous functions (polynomials).
- The set D is closed since it contains all its limit points ($\partial D = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 16\}$ is included in D).
- The set D is bounded since we can find a ball of radius R that contains D (for instance, the ball of radius 5 centered at the origin contains D).

Since f is continuous and D is compact, we can apply the Weierstrass Theorem to conclude that f attains its absolute maximum and minimum values on D . In the interior, we can apply the Hessian criteria to find the critical points of f . The gradient of f is given by:

$$\nabla f(x, y, z) = (2x + 6, 2y - 8, 0).$$

Setting the gradient equal to zero gives us the critical point:

$$2x + 6 = 0 \implies x = -3, \quad 2y - 8 = 0 \implies y = 4, \quad 0 = 0 \implies z \text{ is free.}$$

Thus, the critical points are of the form $(-3, 4, z)$ for any $z \in \mathbb{R}$. However, $(-3, 4, z)$ is not in D since $(-3)^2 + 4^2 = 25 > 16$. Therefore, there are no critical points of f in the interior of D . Now we apply the method of Lagrange multipliers to find the critical points of f on the boundary of D . The constraint is given by:

$$g(x, y, z) = x^2 + y^2 - 16 = 0.$$

The gradient of g is given by:

$$\nabla g(x, y, z) = (2x, 2y, 0).$$

Setting $\nabla f = \lambda \nabla g$ gives us the system of equations:

$$(2x + 6, 2y - 8, 0) = \lambda(2x, 2y, 0) \implies \begin{cases} 2x + 6 &= 2\lambda x \\ 2y - 8 &= 2\lambda y \\ x^2 + y^2 &= 16 \end{cases}$$

From the first two equations, we can express λ in terms of x and y :

$$\lambda = \frac{2x + 6}{2x} = \frac{2y - 8}{2y}.$$

Equating the two expressions for λ gives us:

$$\frac{2x+6}{2x} = \frac{2y-8}{2y} \implies xy+3y = xy-4x \implies 3y = -4x \implies y = -\frac{4}{3}x,$$

if $x \neq 0$ and $y \neq 0$. Substituting $y = -\frac{4}{3}x$ into the constraint equation gives us:

$$x^2 + \left(-\frac{4}{3}x\right)^2 = 16 \implies x^2 + \frac{16}{9}x^2 = 16 \implies \frac{25}{9}x^2 = 16 \implies x^2 = \frac{144}{25} \implies x = \pm\frac{12}{5}.$$

Substituting $x = \pm\frac{12}{5}$ into $y = -\frac{4}{3}x$ gives us:

$$x = \pm\frac{12}{5} \implies y = -\frac{4}{3} \cdot \pm\frac{12}{5} = \mp\frac{16}{5}.$$

Thus, the critical points of f on the boundary of D are $\left(\frac{12}{5}, -\frac{16}{5}, z\right)$ and $\left(-\frac{12}{5}, \frac{16}{5}, z\right)$ for any $z \in \mathbb{R}$. Evaluating f at these critical points gives us:

$$\begin{aligned} f\left(\frac{12}{5}, -\frac{16}{5}, z\right) &= \left(\frac{12}{5}\right)^2 + \left(-\frac{16}{5}\right)^2 + 6 \cdot \frac{12}{5} - 8 \cdot \left(-\frac{16}{5}\right) + 25 \\ &= \frac{144}{25} + \frac{256}{25} + \frac{72}{5} + \frac{128}{5} + 25 \\ &= \frac{400}{25} + \frac{200}{5} + 25 \\ &= 16 + 40 + 25 \\ &= 81, \end{aligned}$$

and

$$\begin{aligned} f\left(-\frac{12}{5}, \frac{16}{5}, z\right) &= \left(-\frac{12}{5}\right)^2 + \left(\frac{16}{5}\right)^2 + 6 \cdot \left(-\frac{12}{5}\right) - 8 \cdot \frac{16}{5} + 25 \\ &= \frac{144}{25} + \frac{256}{25} - \frac{72}{5} - \frac{128}{5} + 25 \\ &= \frac{400}{25} - \frac{200}{5} + 25 \\ &= 16 - 40 + 25 \\ &= 1. \end{aligned}$$

Example: Chapter 2.3, Exercise 6

Compute the absolute maximum and minimum of the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = x^2 + y^2 - 2x - 2y + 2$$

on the set $T = \{(x, y) \in \mathbb{R}^2 : \frac{y}{2} \leq x \leq 3 - \sqrt{2y}, 0 \leq y \leq 2\}$. Again, we can apply the Weierstrass Theorem to conclude that f attains its absolute maximum and minimum values on T since f is continuous and T is compact. In the interior of T , we can apply the Hessian criteria to find the critical points of f . The gradient of f is given by:

$$\nabla f(x, y) = (2x - 2, 2y - 2).$$

Setting the gradient equal to zero gives us the critical point:

$$2x - 2 = 0 \implies x = 1, \quad 2y - 2 = 0 \implies y = 1.$$

Thus, the critical point is $(1, 1)$, and $f(1, 1) = 0$. Let's rewrite f by completing the squares:

$$f(x, y) = (x - 1)^2 + (y - 1)^2.$$

The function f is the distance from the point $(1, 1)$ in the plane. Then, obviously, the minimum value of f is 0, and it is attained at the point $(1, 1)$. The maximum value of f is attained at the point in T that is farthest from $(1, 1)$. The boundary of T consists of three curves:

- The line $x = \frac{y}{2}$ for $0 \leq y \leq 2$.
- The curve $x = 3 - \sqrt{2y}$ for $0 \leq y \leq 2$.
- The line $y = 0$ for $0 \leq x \leq 3$.

Then, the maximum is attained at the point $(3, 0)$

$$f(3, 0) = (3 - 1)^2 + (0 - 1)^2 = 4 + 1 = 5.$$

Example: Chapter 2.3, Exercise 7

Compute the absolute maximum and minimum of the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by:

$$f(x, y, z) = x + 2y + 3z$$

on the set $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 2, x + z = 1\}$, which is the intersection of a cylinder and a plane. We can apply Lagrange multipliers to find the critical points of f on S . The gradient of f is expressed as:

$$\nabla f(x, y, z) = \lambda \nabla g_1(x, y, z) + \mu \nabla g_2(x, y, z),$$

where $g_1(x, y, z) = x^2 + y^2 - 2 = 0$ and $g_2(x, y, z) = x + z - 1 = 0$. The gradients of g_1 and g_2 are given by:

$$\nabla g_1(x, y, z) = (2x, 2y, 0), \quad \nabla g_2(x, y, z) = (1, 0, 1).$$

Setting $\nabla f = \lambda \nabla g_1 + \mu \nabla g_2$ gives us the system of equations:

$$\begin{cases} 1 &= 2\lambda x + \mu \\ 2 &= 2\lambda y \\ 3 &= \mu \end{cases} \implies \begin{cases} 1 &= 2\lambda x + 3 \\ 2 &= 2\lambda y \\ 3 &= \mu \end{cases}$$

From the first equation, we can express λ in terms of x :

$$\lambda = \frac{1 - 3}{2x} = -\frac{1}{x}.$$

Substituting $\lambda = -\frac{1}{x}$ into the second equation gives us:

$$2 = 2\lambda y \implies y = -x.$$

Substituting $y = -x$ into the constraint $g_1(x, y, z) = 0$ gives us:

$$x^2 + (-x)^2 = 2 \implies 2x^2 = 2 \implies x^2 = 1 \implies x = \pm 1.$$

Substituting $x = \pm 1$ into $y = -x$ gives us:

$$x = \pm 1 \implies y = \mp 1.$$

Substituting $x = \pm 1$ into the constraint $g_2(x, y, z) = 0$ gives us:

$$x + z = 1 \implies z = 1 - x.$$

Thus, the critical points of f on S are $(1, -1, 0)$ and $(-1, 1, 2)$. Evaluating f at these critical points gives us:

$$f(1, -1, 0) = 1 + 2(-1) + 3(0) = -1,$$

and

$$f(-1, 1, 2) = -1 + 2(1) + 3(2) = 7.$$

Therefore, the absolute minimum of f on S is -1 and the absolute maximum of f on S is 7 .